A First Course in Abstract Algebra

Section 56: Insolvability of the Quintic

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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56 Insolvability of the Quintic

Exercise 56.1: A radical extension in characteristic two

The polynomial $x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$ and its splitting field is $K = \mathrm{GF}(4)$. A square root of an element of $\mathbb{Z}_2$ cannot generate it: the only elements are 0,1, whose square roots in any extension are uniquely 0,1 in characteristic 2. Nevertheless K is a radical extension. A root α of the polynomial satisfies $\alpha^3 = 1$, and $K = \mathbb{Z}_2(\alpha)$ is obtained by adjoining this cube root of the base-field element 1. A radical need not be a square root or the real positive choice of a root.

Exercise 56.2: An even polynomial of degree eight

Yes in characteristic zero. Substitute $y = x^2$ to get the quartic $ay^4 + by^3 + cy^2 + dy + e$. Every quartic is solvable by radicals, so all four y-roots lie in a radical extension. Adjoining square roots of each of them, one after another, gives a further radical extension containing both square roots of every y-root. These are exactly the roots of the original polynomial. Thus its splitting field is contained in a radical extension.

Exercise 56.3: Ten statements about radicals and solvability

(a) True. This is the definition of solvability by radicals.

(b) True in characteristic zero, by the solvable-Galois-group criterion.

(c) True. The roots of $x^{17} - 5$ are products of a seventeenth root of 5 and powers of a seventeenth root of unity; adjoining these radicals supplies a radical extension. Its group is therefore solvable.

(d) False. Although both are transcendental, they satisfy the polynomial relation $Y^2 - X = 0$ over $\mathbb{Q}$, so they are algebraically dependent.

(e) True. The group is cyclic, hence abelian and solvable.

(f) False. $x^5 - 2$ is solved by adjoining a fifth root of 2 and a fifth root of unity.

(g) True. The quartic formulas, or solvability of subgroups of S_4, prove it.

(h) False. A real cube root of 2 has degree 3 over $\mathbb{Q}$ and cannot belong to a tower of quadratic extensions, whose degree is a power of 2.

(i) False. The cubic $x^3 - x$ has rational roots $-1, 0, 1$, requiring no extension at all.

(j) True. A subnormal series with abelian quotients characterizes solvable groups, linking the group criterion to radical towers.

Exercise 56.4: The quadratic formula over a field

Multiplying $ax^2 + bx + c = 0$ by $4a$ and completing the square gives $(2ax + b)^2 = b^2 - 4ac$. Since $a \neq 0$ and characteristic is not 2, $2a$ is invertible. With $\delta^2 = b^2 - 4ac$, the roots are $(-b + \delta)/(2a)$ and $(-b - \delta)/(2a)$, so they lie in $F(\delta)$. Conversely either root determines $\delta = 2a\alpha + b$, up to sign, so

the field generated by the roots contains δ. Thus the splitting field is $F(\sqrt{b^2-4ac})$, including the trivial-extension cases where the discriminant is already a square or zero.

Exercise 56.5: A biquadratic solved by successive radicals

Put $y = x^2$. First solve $ay^2 + by + c = 0$ by adjoining $\delta = \sqrt{b^2-4ac}$, giving $y_\pm = (-b \pm \delta)/(2a)$. Next adjoin $\sqrt{y_+}$ and $\sqrt{y_-}$. The resulting tower is radical, and contains all four roots $\pm\sqrt{y_+}, \pm\sqrt{y_-}$, with coincidences allowed. Hence the splitting field is contained in this radical extension. The characteristic assumption permits division by $2a$.

Exercise 56.6: Refining a series with abelian factors

Consider one original step $N \trianglelefteq M$ with M/N abelian, and a refined chain $N = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_k = M$. For each j, H_j/N is a subgroup of the abelian group M/N, hence is abelian. By the third isomorphism theorem,

$$H_j/H_{j-1} \cong (H_j/N)/(H_{j-1}/N),$$

which is a quotient of an abelian group and is therefore abelian. Every new step lies within some original step, so this proves the assertion for the whole refinement.

Exercise 56.7: Refining solvable factors to abelian composition factors

For a step $N \trianglelefteq M$ whose quotient M/N is solvable, choose a subnormal series of M/N with abelian factors. Lift its subgroups under the quotient map $M \to M/N$; the correspondence and isomorphism theorems give a subnormal refinement from N to M with those same abelian quotients. Do this for every original step. Because the group is finite, refine further until every nontrivial quotient is simple, obtaining a composition series. Exercise 56.6 ensures the refined quotients remain abelian. Thus all composition factors are simple abelian finite groups, hence cyclic groups of prime order.

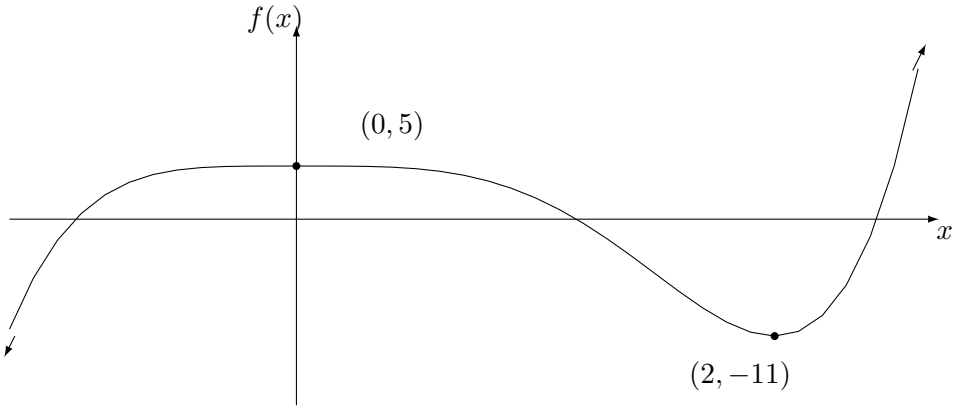
Exercise 56.8: An explicit quintic with no radical solution

(a) Label the letters by $\mathbb{Z}_5$ so the five-cycle is $c : j \mapsto j + 1$. If the transposition is (u, v), its conjugates $c^k(u, v)c^{-k} = (u + k, v + k)$ all lie in H. Regard these as edges joining vertices differing by the nonzero residue $v - u$. Such edges make a connected graph, since every nonzero residue generates the additive group $\mathbb{Z}_5$. Edge transpositions of a connected graph generate all transpositions: along a path, conjugating adjacent edge swaps gives a swap of the path endpoints, by induction on its length. Hence H contains every transposition and therefore $H = S_5$.

(b) Irreducibility of the quintic makes the Galois action on its five roots transitive, so the orbit-stabilizer theorem gives $5 \mid |G|$. Cauchy's theorem gives an element of order 5, which in S_5 is a five-cycle. Complex conjugation exchanges exactly the two nonreal roots and fixes the three real ones, so its permutation is a transposition, not merely an arbitrary element of order 2. Part (a) now gives $G = S_5$.

(c) For $f(x) = 2x^5 - 5x^4 + 5$, Eisenstein at 5 proves irreducibility: 5 divides every nonleading coefficient, 25 does not divide the constant term, and 5 does not divide 2. The derivative is $f'(x) = 10x^3(x - 2)$. Thus f increases on $(-\infty, 0)$, decreases on $(0, 2)$, and increases on $(2, \infty)$.

Its local maximum is $f(0) = 5 > 0$ and its local minimum is $f(2) = -11 < 0$. The odd degree with positive leading coefficient gives limits $-\infty$ and $+\infty$ at the two ends. Therefore continuity and strict monotonicity give exactly one real zero on each of these three intervals, and no others. Its remaining two roots are a nonreal conjugate pair. Here is a sketch, with the two extrema labeled and the curve drawn from actual sampled values; arrows indicate the tails.



By (b) its Galois group is S_5. This group is not solvable: its simple nonabelian normal subgroup A_5 obstructs an abelian-factor series. The solvability theorem therefore shows this quintic is not solvable by radicals over $\mathbb{Q}$.